Evidence of Phonon Drag Effects in $FeSb_2$

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Introduction

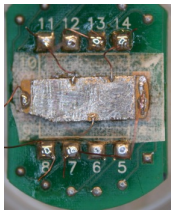
$FeSb_2$ is one of the promising candidates for the Pieltier cooling applications because of colossal Seebeck coefficient at around 10 K. Although a strong electron-electron correlation has been suggested as a possible cause, the complete physics of the origin for such a huge Seebeck coefficient is still missing. Here we aim to analyze the thermoelectric properties of $FeSb_2$ nanocomposites and present evidence of a substantial phonon-drag contribution to the Seebeck coefficient in $FeSb_2$.

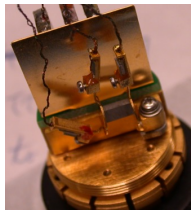
Sample information

Sample code	Hot pressing temperature (°C)	Average grain size (nm)
S-300	300	30
S-400	400	100
S-500	500	350
S-600	600	20,000

Experimental

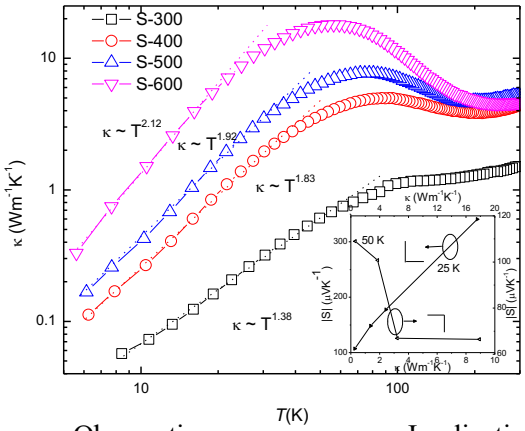
The thermal conductivity (κ), Seebeck coefficient (S), electrical resistivity(ρ) and Hall coefficient (R_H) were all measured in PPMS from Quantum design.





$$\kappa_{total} = \kappa_{lattice} + \kappa_{electronic},\ \kappa_{electronic} = L\sigma T \quad (1)$$

$L = 2.45 \times 10^{-8}\ WK^{-2}\Omega^{-1}$



Observation

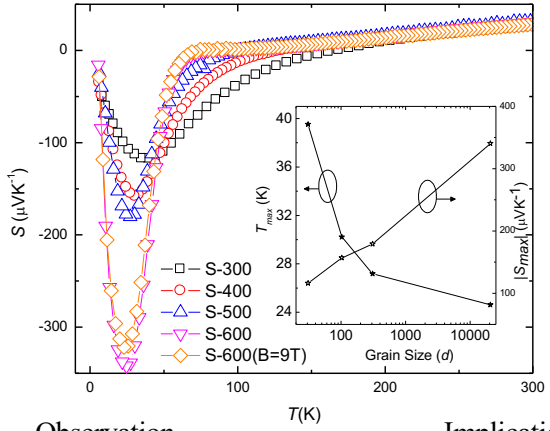
- More than 97% of κ_{total} comes from lattice below 50 K.
- $\kappa \sim T^2$ is nearly obeyed below 50 K for larger grained samples.
- At 25 K, S is proportional to κ.

Implication

- FeSb2 is phonon dominated system at low T
- Electron-phonon coupling
- Phonons affecting S

$$S = S_{diffusion} + S_{phonon} \quad (2)$$

$$S_{diffusion} = \mp 86.2\left[ln\frac{4.7\times10^{15}}{n} + \frac{3}{2}ln\frac{m^*}{m} + \frac{|\Delta E|}{kT} + \frac{3}{2}lnT\right] \quad (3)$$



Observation

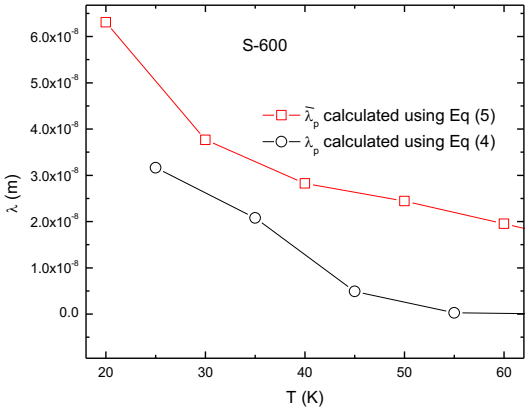
- S is reduced by two roders of magnitudes at low T when compared to the single crystal value
- S max decreases as grain size decreases.
- S max shifts to higher T.

Implication

- Non-electronic origin of S
- Increased scattering of phonons off the grain boundaries suppresses phonon drag contribution
- Size dependent S

$$S_{phonon} = \frac{\beta v_s \lambda_p n e}{\sigma T} \quad (4)$$

$$K_{total} = \frac{1}{3} c_V \widetilde{\lambda_P} v_S \quad (5)$$



- λp and λp were calculated using Eq. (2) - (5).
- The two length scales approximately follow the similar T-dependence
- Similar wavelength phonons might be playing role in both the heat conduction and dragging of electrons.

Conclusion: A substantial contribution to the large S value of coarse-grained $FeSb_2$ nanocomposite sample at low temperature seems to come from phonon-drag effect.